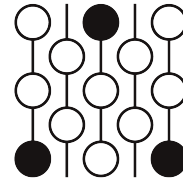


Review of Submission 9A

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

The solution gives a combinatorial formula for the coefficients $c_{(a,1^b)}(n)$ in terms of ordered set partitions.

The proof starts from the diagonal supersymmetry result of [2] to relate the coefficients in question to the Hilbert series of the super coinvariant ring SR_n . It then uses the formula established in [3] for this Hilbert series to express the coefficients using q -Stirling numbers. Finally, some q -series manipulations are then used to obtain the combinatorial formula.

Here are some comments:

- The correct references are used, but they are not cited in the text.
- The presentation is quite dense; all the steps are included in the same proof. I recommend first recalling, in separate theorem environments, Lentfer's result on the universality of the coefficients $c_\lambda(n)$, and Rhoades–Wilson's formula for the Hilbert series. This would make the argument easier to follow.

Correctness. The formula is correct, as is the proof.

Novelty. The two first steps of the proof are standard. The trick used to factorize the series $H_n(q, z)$ (see Eq. (3)) simplifies the last step of the proof. The combinatorial formula is quite nice.

2 Recommendation

I recommend publication after minor revisions.

3 Submission with inline comments

Hook coefficients in the universal Hilbert series of diagonal coinvariants

Problem statement and interpretation

I interpret $R_n^{(m)}$ as the diagonal coinvariant algebra

$$R_n^{(m)} = \mathbb{C}[x_i^{(r)} : 1 \leq i \leq n, 1 \leq r \leq m] / \langle \mathbb{C}[x_i^{(r)}]_+^{\mathfrak{S}_n} \rangle,$$

where $\mathfrak{S}_n$ acts by simultaneously permuting the subscript i in all m sets of variables. Thus the coefficients $c_\lambda(n)$ are the stable multiplicities of the polynomial GL_m -irreducibles $S_\lambda(\mathbb{C}^m)$.

The combinatorial model

For $r, k \geq 1$, let $\text{OP}_{r,k}$ be the set of ordered set partitions $\sigma = (B_1 | \cdots | B_k)$ of $[r]$ into k nonempty blocks. An inversion of σ is a pair (i, j) such that

$$i < j, \quad i = \min B_m, \quad j \in B_\ell, \quad \ell < m.$$

Let $\text{inv}(\sigma)$ be the number of such pairs. When $k = r$, this is the usual inversion number of the permutation obtained by reading the singleton blocks from left to right.

Theorem 1. *Let $\lambda = (a, 1^b)$ be a hook, with $a \geq 1$ and $b \geq 0$. Then $c_{(a, 1^b)}(n)$ is the number of ordered set partitions $\sigma = (B_1 | \cdots | B_k)$ of $[r]$, for some $2 \leq r \leq n$, such that*

$$r - k = b + 1, \quad r \text{ is not a singleton block of } \sigma, \quad \text{inv}(\sigma) = a - 1.$$

Equivalently,

$$\sum_{a \geq 1, b \geq 0} c_{(a, 1^b)}(n) q^{a-1} z^b = \sum_{\substack{2 \leq r \leq n \\ 1 \leq k \leq r-1}} \sum_{\substack{\sigma \in \text{OP}_{r,k} \\ r \text{ not singleton in } \sigma}} q^{\text{inv}(\sigma)} z^{r-k-1}.$$

The empty partition coefficient is $c_\emptyset(n) = 1$.

Proof. We use two standard inputs. First, diagonal supersymmetry says that the universal Schur coefficients for diagonal coinvariants can be read from the coinvariant ring with one bosonic alphabet and one fermionic alphabet,

$$SR_n = (\mathbb{C}[x_1, \dots, x_n] \otimes \bigwedge(\theta_1, \dots, \theta_n)) / \langle (\mathbb{C}[x] \otimes \bigwedge \theta)_+^{\mathfrak{S}_n} \rangle.$$

More explicitly, the supersymmetric Frobenius/character expansion has the same universal multiplicity modules as the purely bosonic expansion; after applying the dimension map to the $\mathfrak{S}_n$ -Frobenius factor one obtains exactly the Hilbert coefficients $c_\lambda(n)$ appearing in the question. Therefore

$$\text{Hilb}(SR_n; q, z) = \sum_{\lambda} c_\lambda(n) s_\lambda(q/z),$$

where $s_\lambda(q/z)$ denotes the super Schur polynomial in one even variable q and one odd variable z . With one even and one odd variable, $s_\lambda(q/z)$ vanishes unless λ lies in the $(1, 1)$ -hook, i.e. unless λ is a hook, and

$$s_{(a, 1^b)}(q/z) = q^a z^b + q^{a-1} z^{b+1} = q^{a-1} z^b (q + z).$$

Add a reference for this formula.

Thus

$$\text{Hilb}(SR_n; q, z) = 1 + (q + z) \sum_{a \geq 1, b \geq 0} c_{(a, 1^b)}(n) q^{a-1} z^b. \quad (1)$$

Second, the Hilbert series of SR_n , due to Rhoades–Wilson, is

$$H_n(q, z) := \text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} [k]_q! \text{Stir}_q(n, k), \quad (2)$$

where $[k]_q = 1 + q + \dots + q^{k-1}$, $[k]_q! = [k]_q [k-1]_q \dots [1]_q$, and

$$\text{Stir}_q(r, k) = \text{Stir}_q(r-1, k-1) + [k]_q \text{Stir}_q(r-1, k).$$

The boundary conditions are $\text{Stir}_q(0, 0) = 1$, $\text{Stir}_q(r, 0) = 0$ for $r > 0$, and $\text{Stir}_q(r, k) = 0$ outside $0 \leq k \leq r$. Put $F_{r,k} = [k]_q! \text{Stir}_q(r, k)$. Then $F_{r,k} = 0$ outside $1 \leq k \leq r$, except for the harmless convention $F_{0,0} = 1$, and

$$F_{r,k} = [k]_q F_{r-1,k-1} + [k]_q F_{r-1,k}.$$

Using $[k+1]_q = 1 + q[k]_q$ in (2), we get, for $n \geq 2$,

$$\begin{aligned} H_n(q, z) &= \sum_{k=1}^n z^{n-k} ([k]_q F_{n-1,k-1} + [k]_q F_{n-1,k}) \\ &= H_{n-1}(q, z) + (q + z) \sum_{k=1}^{n-1} [k]_q F_{n-1,k} z^{n-1-k}. \end{aligned}$$

Since $H_1(q, z) = 1$, iteration gives

$$\frac{H_n(q, z) - 1}{q + z} = \sum_{r=1}^{n-1} \sum_{k=1}^r [k]_q F_{r,k} z^{r-k}. \quad (3)$$

It remains to identify the right side of (3). The polynomial $F_{r,k}$ is the inversion enumerator of ordered set partitions:

$$F_{r,k} = \sum_{\tau \in \text{OP}_{r,k}} q^{\text{inv}(\tau)}. \quad (4)$$

Indeed, both sides satisfy the same recurrence. To see this directly, insert the largest letter r into an ordered partition of $[r-1]$. To obtain k final blocks, either insert r as a singleton block into one of the k possible final block positions, or insert r into one of the k existing blocks. If the final position is i from the left, the new inversions involving r are exactly the minima of the blocks to its right, so their number is $k-i$. Summing over $i = 1, \dots, k$ contributes $[k]_q$ in each of the two cases, giving the recurrence above and the initial condition.

Now restrict this insertion argument to insertions of $r+1$ into an existing block of an ordered partition $\tau \in \text{OP}_{r,k}$. The result is precisely an ordered partition $\sigma \in \text{OP}_{r+1,k}$ in which $r+1$ is not a singleton, and the extra inversion contribution is again one of $0, 1, \dots, k-1$. Hence

$$[k]_q F_{r,k} = \sum_{\substack{\sigma \in \text{OP}_{r+1,k} \\ r+1 \text{ not singleton}}} q^{\text{inv}(\sigma)}.$$

Substituting this into (3), and replacing $r + 1$ by r , gives exactly

$$\sum_{a \geq 1, b \geq 0} c_{(a,1^b)}(n) q^{a-1} z^b = \sum_{\substack{2 \leq r \leq n \\ 1 \leq k \leq r-1}} \sum_{\substack{\sigma \in \text{OP}_{r,k} \\ r \text{ not singleton}}} q^{\text{inv}(\sigma)} z^{r-k-1}.$$

Extracting the coefficient of $q^{a-1} z^b$ proves the claimed interpretation. □

References

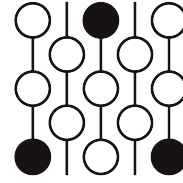
- [1] F. Bergeron, *Multivariate diagonal coinvariant spaces for complex reflection groups*, Adv. Math. 239 (2013), 97–108.
- [2] J. Lentfer, *Diagonal supersymmetry for coinvariant rings*, arXiv:2505.14885 (2025).
- [3] B. Rhoades and A. T. Wilson, *The Hilbert series of the superspace coinvariant ring*, Forum Math. Pi 12 (2024), e16.

Review of Submission 9A

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

The proof is quite elegant and short, with the main points being:

1. Use diagonal supersymmetry and the structure of super schur functions for hook shapes to show the Hilbert series of the superspace ¹ coinvariant ring is of the form $H_n = 1 + (q + z) \sum_{a \geq 1, b \geq 0} c_{(a, 1^b)}(n) q^{a-1} z^b$.
2. Use the Rhoades–Wilson formula for H_n , which is in terms of q -Stirling numbers, and classical recursions of q -Stirling numbers to show that H_n satisfies $\frac{H_n - 1}{q + z} = \sum_{r=1}^{n-1} \sum_{k=1}^r [k]_q F_{r,k} z^{r-k}$, where $F_{r,k} = [k]_q! \text{Stir}_q(r, k)$.
3. Show that $[k]_q F_{r,k}$ is the generating function tracking inversions of ordered set partitions of $\{1, 2, \dots, r+1\}$ into k parts with $r+1$ not in its own block.
4. Finally, to prove the claimed interpretation, simply substitute (3) into (2), then extract the coefficient of $q^{a-1} z^b$, which by (1), matches $c_{(a, 1^b)}(n)$.

Correctness. This proof seems to be correct, as far as I’m able to tell. There are some incorrect (or confusing) superfluous details, and there might be a missing (but very easy) edge case (see inline comments). Otherwise, the argument seems true.

Novelty. In some sense, I find this combinatorial interpretation the most natural and elegant of the four model solutions – it is certainly the easiest formula to compute examples with! The proof is also much shorter than the others, and also much shorter than the human solution. However, given that the combinatorics of the superspace ring has a lot to do with ordered set partitions of $\{1, 2, \dots, n\}$, it in some sense feels “unnatural” that this interpretation is given in terms of ordered set partitions of $\{1, \dots, r\}$ as r varies, rather than just for $r = n$. (The human solution is the only one whose combinatorial interpretation is in terms of special ordered set partitions of $\{1, 2, \dots, n\}$.)

¹one commuting variable, one anticommuting variable

Attributions. Although this submission includes all the papers it used in its final bibliography, it is missing several important in-text citations. It says some statements are “standard” which are actually the main results of these recent papers. The rare in-text citation does not give explicit theorem statements, putting extra work on the reader.

Exposition. The claim is proven all in one theorem. It would be nicer to read if it was broken into several lemmas. There are also a couple of unnecessary/irrelevant comments which do not make sense to me, though this doesn’t actually impact the argument.

2 Recommendation

I would categorize this solution as publishable with minor revisions.

3 Submission with inline comments

Hook coefficients in the universal Hilbert series of diagonal coinvariants

Problem statement and interpretation

I interpret $R_n^{(m)}$ as the diagonal coinvariant algebra

$$R_n^{(m)} = \mathbb{C}[x_i^{(r)} : 1 \leq i \leq n, 1 \leq r \leq m] / \langle \mathbb{C}[x_i^{(r)}]_+^{\mathfrak{S}_n} \rangle,$$

where $\mathfrak{S}_n$ acts by simultaneously permuting the subscript i in all m sets of variables. Thus the coefficients $c_\lambda(n)$ are the stable multiplicities of the polynomial GL_m -irreducibles $S_\lambda(\mathbb{C}^m)$.

Reviewer 2

Although this does not actually matter to the argument, I am a bit confused by the meaning of the last sentence. This would make more sense to me if the coefficients $c_\lambda(n)$ were arising by somehow looking at the character of an action of GL_m on $R_n^{(m)}$, but they are instead arising within the *Hilbert series* of $R_n^{(m)}$?

The combinatorial model

For $r, k \geq 1$, let $\text{OP}_{r,k}$ be the set of ordered set partitions $\sigma = (B_1 | \cdots | B_k)$ of $[r]$ into k nonempty blocks. An inversion of σ is a pair (i, j) such that

$$i < j, \quad i = \min B_m, \quad j \in B_\ell, \quad \ell < m.$$

Let $\text{inv}(\sigma)$ be the number of such pairs. When $k = r$, this is the usual inversion number of the permutation obtained by reading the singleton blocks from left to right.

Theorem 1. Let $\lambda = (a, 1^b)$ be a hook, with $a \geq 1$ and $b \geq 0$. Then $c_{(a, 1^b)}(n)$ is the number of ordered set partitions $\sigma = (B_1 | \cdots | B_k)$ of $[r]$, for some $2 \leq r \leq n$, such that

$$r - k = b + 1, \quad r \text{ is not a singleton block of } \sigma, \quad \text{inv}(\sigma) = a - 1.$$

Equivalently,

$$\sum_{a \geq 1, b \geq 0} c_{(a, 1^b)}(n) q^{a-1} z^b = \sum_{\substack{2 \leq r \leq n \\ 1 \leq k \leq r-1}} \sum_{\substack{\sigma \in \text{OP}_{r,k} \\ r \text{ not singleton in } \sigma}} q^{\text{inv}(\sigma)} z^{r-k-1}.$$

The empty partition coefficient is $c_{\emptyset}(n) = 1$.

Proof. We use two standard inputs. First, diagonal supersymmetry says

Reviewer 2

Should cite J. Lentfer! This is a new paper, not a “standard input.”

that the universal Schur coefficients for diagonal coinvariants can be read from the coinvariant ring with one bosonic alphabet and one fermionic alphabet,

$$SR_n = (\mathbb{C}[x_1, \dots, x_n] \otimes \bigwedge(\theta_1, \dots, \theta_n)) / ((\mathbb{C}[x] \otimes \bigwedge \theta)_+^{\mathfrak{S}_n}).$$

More explicitly, the supersymmetric Frobenius/character expansion has the same universal multiplicity modules as the purely bosonic expansion; after applying the dimension map to the $\mathfrak{S}_n$ -Frobenius factor one obtains exactly the Hilbert coefficients $c_\lambda(n)$ appearing in the question.

Reviewer 2

The next conclusion is correct, but I am confused by this explanation in the previous sentence – what is meant by the “purely bosonic expansion”?

Therefore

$$\text{Hilb}(SR_n; q, z) = \sum_{\lambda} c_\lambda(n) s_\lambda(q/z),$$

where $s_\lambda(q/z)$ denotes the super Schur polynomial in one even variable q and one odd variable z . With one even and one odd variable, $s_\lambda(q/z)$ vanishes unless λ lies in the $(1, 1)$ -hook

Reviewer 2

What do you mean λ lies in the $(1, 1)$ -hook? Also, there should be a condition on a here (≥ 1).

, i.e. unless λ is a hook, and

$$s_{(a, 1^b)}(q/z) = q^a z^b + q^{a-1} z^{b+1} = q^{a-1} z^b (q + z).$$

Thus

$$\text{Hilb}(SR_n; q, z) = 1 + (q + z) \sum_{a \geq 1, b \geq 0} c_{(a, 1^b)}(n) q^{a-1} z^b. \quad (1)$$

Second, the Hilbert series of SR_n , due to Rhoades–Wilson, is

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where $[k]_q = 1 + q + \dots + q^{k-1}$, $[k]!_q = [k]_q [k-1]_q \dots [1]_q$, and

$$\text{Stir}_q(r, k) = \text{Stir}_q(r-1, k-1) + [k]_q \text{Stir}_q(r-1, k).$$

Reviewer 2

Missing $k = 0$ term in Hilbert series? But I guess this only matters in the edge case that $n = 0$? This trivial edge case should probably be addressed.

The boundary conditions

Reviewer 2

Isn't the third set of boundary conditions (that $\text{Stir}_q(r, k) = 0$ outside $k \leq r$ is superfluous? Doesn't it follow from the first two sets.

are $\text{Stir}_q(0, 0) = 1$, $\text{Stir}_q(r, 0) = 0$ for $r > 0$, and $\text{Stir}_q(r, k) = 0$ outside $0 \leq k \leq r$. Put $F_{r,k} = [k]!_q \text{Stir}_q(r, k)$. Then $F_{r,k} = 0$ outside $1 \leq k \leq r$, except for the harmless convention $F_{0,0} = 1$, and

$$F_{r,k} = [k]_q F_{r-1,k-1} + [k]_q F_{r-1,k}.$$

Using $[k+1]_q = 1 + q[k]_q$ in (2), we get, for $n \geq 2$,

$$\begin{aligned} H_n(q, z) &= \sum_{k=1}^n z^{n-k} ([k]_q F_{n-1,k-1} + [k]_q F_{n-1,k}) \\ &= H_{n-1}(q, z) + (q + z) \sum_{k=1}^{n-1} [k]_q F_{n-1,k} z^{n-1-k}. \end{aligned}$$

Since $H_1(q, z) = 1$, iteration gives

$$\frac{H_n(q, z) - 1}{q + z} = \sum_{r=1}^{n-1} \sum_{k=1}^r [k]_q F_{r,k} z^{r-k}. \quad (3)$$

It remains to identify the right side of (3). The polynomial $F_{r,k}$ is the inversion enumerator of ordered set partitions:

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Indeed, both sides satisfy the same recurrence. To see this directly, insert the largest letter r into an ordered partition of $[r-1]$. To obtain k final blocks, either insert r as a singleton block into one of the k possible final block positions, or insert r into one of the k existing blocks. If the final position is i from the left, the new inversions involving r are exactly the minima of the blocks to its right, so their number is $k-i$. Summing over $i = 1, \dots, k$ contributes $[k]_q$ in each of the two cases, giving the recurrence above and the initial condition.

Now restrict this insertion argument to insertions of $r+1$ into an existing block of an ordered partition $\tau \in \text{OP}_{r,k}$. The result is precisely an ordered partition $\sigma \in \text{OP}_{r+1,k}$ in which $r+1$ is not a singleton, and the extra inversion contribution is again one of $0, 1, \dots, k-1$. Hence

$$[k]_q F_{r,k} = \sum_{\substack{\sigma \in \text{OP}_{r+1,k} \\ r+1 \text{ not singleton}}} q^{\text{inv}(\sigma)}.$$

Substituting this into (3), and replacing $r + 1$ by r , gives exactly

$$\sum_{a \geq 1, b \geq 0} c_{(a,1^b)}(n) q^{a-1} z^b = \sum_{\substack{2 \leq r \leq n \\ 1 \leq k \leq r-1}} \sum_{\substack{\sigma \in \text{OP}_{r,k} \\ r \text{ not singleton}}} q^{\text{inv}(\sigma)} z^{r-k-1}.$$

Extracting the coefficient of $q^{a-1} z^b$ proves the claimed interpretation. □

References

- [1] F. Bergeron, *Multivariate diagonal coinvariant spaces for complex reflection groups*, Adv. Math. 239 (2013), 97–108.
- [2] J. Lentfer, *Diagonal supersymmetry for coinvariant rings*, arXiv:2505.14885 (2025).
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